

PIAN ROSA, Italy

WMO: 160520

Lat: 45.935N

Lon: 7.706E

Elev: 3488

StdP: 65.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -21.9	(d) -32.1	(e) 0.3	(f) -11.9	(g) -30.2	(h) 0.4	(i) -9.9	(j) 26.2	(k) -14.5	(l) 24.7	(m) -14.0	(n) 11.4	(o) 60	(p) 1.070

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.4	(c) 9.3	(d) 4.4	(e) 8.1	(f) 3.9	(g) 7.1	(h) 3.2	(i) 5.9	(j) 8.2	(k) 5.0	(l) 7.1	(m) 4.1	(n) 6.2	(o) 2.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.8	8.2	6.8	3.8	7.7	5.9	2.8	7.1	5.1	28.4	8.2	26.0	7.0	23.8	6.2	9.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.2	15.3	-25.7	11.2	2.5	2.2	-27.5	12.8	-28.9	14.1	-30.3	15.3	-32.1	16.9
			-25.4	7.3	2.2	1.0	-27.0	8.0	-28.3	8.6	-29.6	9.1	-31.2	9.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.0	-11.7	-11.3	-10.3	-7.3	-3.6	-0.1	2.6	2.4	-0.5	-3.0	-7.9	-10.2
	DBStd	6.51	4.67	4.67	4.43	3.52	3.40	3.94	3.11	3.02	3.48	3.57	4.84	4.64
	HDD10.0	5488	672	598	630	519	423	302	228	235	316	402	538	626
	HDD18.3	8530	931	831	888	769	682	552	486	493	566	660	788	884
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	6.0	8.2	7.1	6.8	5.9	5.5	4.8	4.5	4.1	4.7	5.6	7.0	7.8
(15) Precipitation	PrecAvg	911	55	55	59	84	113	87	72	73	79	88	86	60
	PrecMax	1261	123	131	124	163	216	173	167	142	205	213	217	173
	PrecMin	648	10	9	7	28	56	40	25	38	17	32	21	6
	PrecStd	149	32	34	31	40	45	34	29	26	48	47	56	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	0.8	0.2	2.0	6.9	10.4	11.1	10.5	8.1	4.2	2.1	-0.8
		MCWB	-6.5	-4.6	-4.9	-2.3	1.8	3.4	5.6	5.3	3.4	-0.6	-4.4	-6.4
	2%	DB	-2.9	-1.7	-2.0	0.0	4.4	8.2	9.5	8.9	6.4	3.2	0.2	-2.3
		MCWB	-7.0	-6.1	-5.9	-3.5	0.1	3.2	5.0	4.7	2.2	-1.5	-5.0	-6.8
	5%	DB	-4.4	-3.8	-3.7	-1.3	2.4	6.9	8.2	7.7	5.2	2.2	-1.0	-3.5
		MCWB	-7.7	-7.7	-7.1	-4.4	-1.1	3.0	4.1	3.8	1.0	-2.1	-5.2	-7.2
	10%	DB	-5.8	-5.2	-4.9	-2.7	1.0	5.2	7.1	6.6	4.1	1.3	-2.1	-4.7
		MCWB	-8.6	-8.6	-7.9	-5.2	-2.1	1.9	3.3	3.0	0.4	-2.9	-5.7	-7.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.3	-4.0	-2.9	-1.1	2.8	6.5	7.1	7.1	5.2	1.9	-1.9	-3.7
		MCDB	-3.4	-0.9	-1.1	0.4	6.0	8.4	9.3	9.0	6.9	3.0	-0.4	-3.4
	2%	WB	-5.7	-5.7	-5.1	-2.5	1.0	4.7	6.1	5.8	3.7	0.6	-3.1	-5.1
		MCDB	-4.1	-3.0	-3.3	-1.0	3.3	6.9	8.5	7.9	5.4	1.9	-2.1	-4.2
	5%	WB	-7.0	-7.0	-6.2	-3.7	-0.4	3.6	5.1	4.8	2.6	-0.4	-4.0	-6.2
		MCDB	-5.2	-4.9	-4.3	-2.2	1.8	5.8	7.5	6.7	4.3	1.0	-2.4	-4.9
	10%	WB	-8.1	-8.1	-7.2	-4.8	-1.5	2.5	4.1	3.8	1.6	-1.4	-4.9	-7.3
		MCDB	-6.1	-5.9	-5.5	-3.4	0.3	4.7	6.4	5.8	3.2	0.1	-3.1	-5.5
(35) Mean Daily Temperature Range	MDBR		4.6	4.8	4.9	5.0	5.0	5.2	5.4	4.9	4.3	4.1	4.2	4.5
	5% DB	MCDBR	4.8	5.3	5.2	5.9	6.3	6.6	6.5	5.6	4.9	4.3	4.3	4.3
		MCWBR	3.9	4.3	4.4	5.3	5.8	6.3	6.3	6.2	6.4	4.8	3.7	3.7
	5% WB	MCDBR	4.7	5.5	5.0	5.8	5.9	6.3	6.2	5.4	4.7	3.9	3.8	4.5
		MCWBR	4.7	5.3	4.8	5.6	5.7	6.4	6.4	6.3	6.3	5.1	4.2	4.8
(40) Clear-Sky Solar Irradiance	taub		0.182	0.196	0.223	0.251	0.263	0.270	0.268	0.261	0.248	0.231	0.204	0.180
	taud		2.429	2.397	2.376	2.422	2.492	2.502	2.505	2.538	2.571	2.597	2.545	2.449
	Ebn at Noon		991	1026	1029	1016	1006	995	993	993	988	970	953	959
	Edh at Noon		65	83	101	107	104	103	102	94	83	69	58	56
(44) All-Sky Solar Radiation	RadAvg		1.53	2.40	3.53	4.42	5.07	6.01	6.06	5.26	4.08	2.62	1.54	1.27
	RadStd		0.18	0.31	0.47	0.57	0.43	0.49	0.47	0.39	0.33	0.30	0.23	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (102 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.70	+0.40	+0.30	N/A	N/A	+56	+16

Nomenclature: See separate page